Roger Tawfik

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Education

University of Massachusetts Amherst **Amherst, MA**
Bachelor of Science in Computer Engineering *Expected May 2028*

- **Academic Distinctions:** GPA: 3.804/4.00 | Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- **Relevant Coursework:** Continuous-Time Signals & Systems, Embedded Systems, Circuits & Electronics, Systems Programming, Hardware Organization & Digital Design, Probability & Statistics, Advanced Programming

Work Experience

Computer Vision Research Engineer **Jun 2026 - Present**
I-SENSE, Florida Atlantic University *Boca Raton, FL*
Python, Kotlin, Swift, OpenCV, MediaPipe Tasks, YOLOv8, TFLite, Core ML, LangChain, MeTRAbs, SQLite, Git
Outstanding Research Award, 2026 NSF REU Cohort, FAU I-SENSE | Project Page

- Built an end-to-end **clinical gait-video quality control (QC) system** in Python and native Android/iOS using OpenCV, MediaPipe, and YOLOv8 to deliver on-device pass-or-re-record feedback before downstream analysis
- Validated one QC specification across three independent implementations (desktop Python, native Kotlin Android, native Swift iOS) on a **shared 120-video study corpus**, using per-check reconciliation tooling to isolate platform-specific decision-boundary mismatches; set the blur and brightness gates from **Zenomat instrumented-walkway ground truth** on 18 paired clinical clips, with repeat-run determinism testing
- Profiled detection to its bottleneck: per-frame model-call overhead, ∼85% of stage time, batched YOLOv8 with a custom CPU track linker for a ∼**5x speedup**, proving **zero verdict change** on all 18 benchmark clips and every check
- Built **regression tests, structured logging**, SQLite aggregation database, per-check reconciliation, and frame-level diagnostic tooling to isolate tracking defects and support the ongoing publication analysis

Personal Projects

AI Nutrition Tracker App | *TypeScript, React, Supabase, Vite/PWA* **Jul 2026 - Present**

- Built a mobile-first nutrition tracker supporting **natural-language, photo, and barcode food logging**, adaptive calorie targets, nutrient-gap recommendations, and daily AI-generated coaching insights for low-friction tracking
- Architected a shared ingestion workflow with Supabase authentication, Postgres, and edge functions, converting text, photo, and barcode inputs into **editable nutrition records** for user review and correction before saving

Go Fish Game | *C, Ubuntu Linux, GCC, Make* **Oct 2026 - Nov 2026**

- Implemented a command-line Go Fish game in C on Ubuntu Linux, with deck shuffling, human-versus-computer turns, rank validation, book detection, and replay controls across separate source and header files
- Managed player hands with **dynamically allocated linked lists**, implementing pointer-based card insertion, removal, and transfer with **malloc/free cleanup**, and compiling and linking the program with GCC and Make

Gym Form Trainer | *Python, C, OpenCV, MediaPipe, Arduino* **Dec 2025 - May 2026**

- Built an exercise form trainer with MediaPipe Pose and OpenCV, computing smoothed joint angles to score squats, bicep curls, and push-ups against exercise-specific thresholds while counting repetitions through a **state machine**
- Integrated the Python vision pipeline with **C firmware over bidirectional serial communication**, driving LED score feedback and rep-count buzzer alerts while accepting hardware-button exercise controls during live workouts

Algorithmic Trading Analysis System | *Python, Pine Script, SQLite, LightGBM* **Feb 2026 - Apr 2026**

- Engineered an automated market-analysis system in Python to screen **4,622 stocks**, combine multi-factor technical signals into quantitative scores, and identify breakout candidates for large-scale historical evaluation and backtesting
- Built a trigger-aware historical labeling pipeline over **662,678 ticker-date observations**, using LightGBM, **time-ordered validation**, SHAP analysis, isotonic calibration, SQLite caching, and Pine Script backtesting

Technical Skills

Languages: Python, C, C++, TypeScript, Java, Kotlin, Swift, SQL, ARM/AVR Assembly, MATLAB, Pine Script
AI & Computer Vision: OpenCV, MediaPipe, YOLOv8, TFLite, Core ML, LightGBM, LangChain, Random Forest
Web & Mobile: React, Supabase, REST APIs, Android Development, SwiftUI, Vite/PWA, PostgreSQL, Edge Functions
Systems & Tools: Ubuntu Linux/Unix, Bash, GCC, Make, Git, Arduino, Raspberry Pi, Concurrent Programming, Network Programming, Computer Architecture, Inter-Process Communication, Digital Logic, Network Security, Fuzzing, Pandas, NumPy
Testing & Validation: Regression Testing, Cross-Platform Validation, Performance Profiling, Benchmarking, Structured Logging, Reproducible Experiments